



Intellect Aqua Private Limited
GROUP OF PRIME AQUA TECH

Water Treatment
RO - SOFTENER

Waste Water Treatment
STP - ETP - UF

Renewable Energy
BIO GAS



CLIENT

M/s. ANANYA SHELTERS PRIVATE LIMITED,
COIMBATORE.
PO SITE : HOTEL@ KAVUNDAMPALAYAM

PROJECT : STP 50 KLD

TECHNOLOGY : MBR - CIVIL

PROPOSAL NO : IAPL/QT/ANN/STP/9665R5

DATE : 25.04.2026

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No.70, RR Layout, New Siddhapudur, Siddhapudur, Coimbatore – 641 044, India.





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APRIL 25, 2026

To

M/s. ANANYA SHELTERS PRIVATE LIMITED,
COIMBATORE

Kind Attn: Mr. Praveen R, Associate Director

Dear Sir,

Sub: Techno-Commercial Proposal for 50 KLD MBR Based Civil STP - Reg.

With reference to your enquiry, we are pleased to submit our **Techno-Commercial Proposal** for the **design, engineering, construction of civil structures, supply, erection, testing, and commissioning** of a **50 KLD Sewage Treatment Plant (STP)** based on **MBR (Membrane Bioreactor)** technology, proposed to be installed at your site.

Intellect Aqua Private Limited has proven expertise in executing **compact, high-efficiency MBR-based STPs with integrated civil and electro-mechanical works**, designed to meet **CPCB / TNPCB discharge norms** and site-specific requirements. The proposed system ensures **superior treated water quality, minimal footprint, odour-free operation, low sludge generation, and consistent performance**, making it ideal for reuse applications such as flushing, gardening, and utility purposes.

Our solution is engineered with a focus on **robust civil construction, optimized hydraulic design, reliable MBR membrane performance, automation-ready operation, and long-term operational stability**, resulting in **lower life-cycle costs and ease of maintenance**.

We are confident that our **end-to-end execution capability, quality workmanship, and competitive commercial offering** will add measurable value to your project. We trust that our proposal will meet your expectations and look forward to the opportunity of receiving your valued order.

For any further clarification or for detailed technical or commercial discussions, our team would be pleased to meet you at your convenience.

Thanking you and assuring you of our **best services, prompt support, and a long-term professional partnership**.

Thanking you,
Yours Truly,

For Intellect Aqua Private Limited

Rtn. BOOPATHI S.P

Chief Managing Director

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ABOUT INTELLECT - STP

Intellect Aqua proposes a state-of-the-art **Sewage Treatment Plant (STP)** based on **Membrane Bio-Reactor (MBR) technology**, designed to achieve high-quality treated water suitable for reuse and safe discharge as per Pollution Control Board norms. The MBR system integrates biological treatment with advanced membrane filtration, ensuring effective removal of suspended solids, organic matter, and pathogens. This compact, fully automatic system provides consistent performance with minimal operator attention, low footprint, and reduced sludge generation. Our design ensures reliable operation, ease of maintenance, and long-term efficiency—making it an ideal solution for residential, commercial, and industrial wastewater treatment applications.

Key Features of MBR Sewage Treatment Plant

- **Advanced Membrane Filtration:** Combines biological treatment and membrane separation for superior effluent quality.
- **Compact & Space Saving Design:** Requires significantly less area compared to conventional systems.
- **High Treated Water Quality:** Produces clear, odour-free, and disinfected water suitable for reuse in gardening, flushing, and cooling towers.
- **Automatic Operation:** PLC-based or fully automatic operation ensures minimal manual intervention.
- **Low Sludge Generation:** Optimized biological process reduces excess sludge production.
- **Modular Construction:** Easy to install, expand, and relocate as per site requirements.
- **Odour-Free & Silent Operation:** Enclosed process eliminates foul odour and noise nuisance.
- **Durable Materials:** All tanks, piping, and membranes are corrosion-resistant for long service life.
- **Energy Efficient:** Designed for low power consumption and cost-effective operation.
- **Easy Maintenance:** Simple cleaning and backwash process ensures long membrane life.

Benefits of MBR Technology

- **Consistent Effluent Quality:** Meets and exceeds **Pollution Control Board norms** even under fluctuating load conditions.
- **Water Reuse & Conservation:** Treated water can be reused, reducing freshwater dependency.
- **Reduced Footprint:** Ideal for projects with limited space such as apartments, hotels, hospitals, and industries.
- **Environmentally Sustainable:** Promotes eco-friendly wastewater management and resource recovery.
- **Long-Term Reliability:** Proven technology with stable performance and reduced downtime.

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DESIGN PARAMETERS:

A. Flow parameters	
Nature of waste water	Sewage
Flow rate (KLD)	50 KLD
Operating hours	20 Hours

B. Raw waste water parameters	
pH	7.0 - 8.5
COD	≤ 600ppm
BOD – (3 days @ 27 °C)	≤ 300 ppm
TSS	≤ 100 ppm
Oil & Grease	≤ 10 ppm

C. Treated waste water parameters	
pH	7.0 - 8.5
COD	≤ 30 ppm
BOD – (3 days @ 27 °C)	≤10 ppm
TSS	≤ 1 ppm
Oil & Grease	≤ Traces

Note: Above treated water parameters are for using treated water in flushing

ASSUMPTIONS:

1. The plant is designed to operate at +/- 10 % variation in raw wastewater parameter.
2. No other parameters other than mentioned above is present in the raw waste water which is beyond Pollution Control Norms and hazardous to micro-organisms.
3. Treated water quality will be achieved if the inlet raw water quality is as per the raw water quality mentioned as well as no other pollutant than the mentioned, is present or exceeds the limits or which is hazardous in nature, which otherwise may affect the biological treatment process



PROCESS DESIGN DETAILS: -

Screen Chamber: -

Normal flow designed	:	2.5m ³ /hr.
No. of Screen Chamber	:	One
Angle of Inclination	:	45 ⁰
Size Of the screen chamber	:	2.3m X 1.1m x 0.6m
MOC of screen chamber	:	RCC

Oil Trap Chamber: -

Normal flow designed	:	2.5 m ³ /hr.
Size Of the Tank	:	2.3m X 1.7m x 0.6m
MOC of the Tank	:	RCC

Pre-Collection Tank:

Normal Flow Designed	:	2.5 m ³ /hr.
No. of Tanks	:	ONE
Volume of the Tank	:	10 m ³
Size of the Tank	:	3.7 m x 2.5 m x 1.4 m
MOC	:	RCC

Flocculation Tank:

Normal Flow Designed	:	2.5 m ³ /hr.
No. of Tanks	:	ONE
Size of the Tank	:	2.0 m x 1.2 m x 0.3 m
MOC	:	RCC



Equalization Tank:

Normal Flow Designed	:	2.5 m ³ /hr.
No. of Tanks	:	ONE
Retention Time	:	10 Hrs.
Volume of the Tank	:	25 m ³
Size of the Tank	:	4.2 m x 3.5 m x 2.2 m
MOC	:	RCC

ANOXIC Tank:

Normal Flow Designed	:	2.5m ³ /hr.
No. of Tanks	:	ONE
Retention Time	:	4 Hrs.
Volume of the Tank	:	10 m ³
Size of the Tank	:	4.2 m x 2.3 m x 2.2 m
MOC	:	RCC

Aeration Tank:

Normal Flow Designed	:	2.5m ³ /hr.
No. of Tanks	:	ONE
Retention Time	:	8hrs
Volume of the Tank	:	20m ³
Size of the Tank	:	2.6 m x 4.8 m x 2.2 m
MOC	:	RCC





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MBR Tank:

Normal Flow Designed	:	2.5m ³ /hr.
No. of Tanks	:	ONE
Retention Time	:	4hrs
Volume of the Tank	:	10m ³
Size of the Tank	:	2.6 m x 2.1 m x 2.2 m
MOC	:	RCC

Treated Water Tank: - (Optional)

Normal Flow Designed	:	2.5m ³ /hr.
Retention Time	:	4hrs
Volume of the Tank	:	10m ³
Size of the Tank	:	3.3 m x 1.25 m x 2.2m
MOC	:	RCC/HDPE

Sludge Holding Tank: - (Client Scope)

Normal Flow Designed	:	2.5m ³ /hr.
Volume of the Tank	:	5m ³
MOC	:	RCC/ HDPE

CIP Tank: - (Client Scope)

No Of Tank	:	1 No
Volume of the Tank	:	1000m ³
MOC	:	RCC/ HDPE





OUR SCOPE OF WORK & SPECIFICATIONS FOR SUPPLYING EQUIPMENTS: -

1. Bar Screen (COARSE)

DESCRIPTION	
Quantity	1 No
Spacing	15mm
Type	Mechanical Auto Rake
MOC	SS 304
Gear Box	Transtech
RPM	14 RPM

2. Rotary Drum Screen

DESCRIPTION	
Quantity	1 No
Spacing	1mm
Type	Rotary Drum (OUT TO IN)
MOC	SS 304
Gear Box	Transtech
RPM	14 RPM

3.Oil Skimmer

DESCRIPTION	
Quantity	2 Nos
Type	Belt Type
MOC	SS 304
Gear Box	WORMGEAR
RPM	14 RPM

4. Raw Water Pump (Pre-Collection tank to Flocculation Chamber)

DESCRIPTION	
Quantity	2 Nos (1W+1S)
Motor	0.5HP
Head	10 m
Type	Submersible Pump
Make	Kirloskar



5.Raw Water Pump (Collection Tank to Anoxic Tank)

DESCRIPTION	
Quantity	2 Nos (1W+1S)
Motor	1HP
Head	10 m
Type	Submersible Pump
Make	Kirloskar

6. Agitator for Anoxic Tank

DESCRIPTION	
Quantity	1 No.
Make	IAPL
Type	Submersible
Power	1.1 KW

7. Air Blower

DESCRIPTION	
Quantity	2Nos (1W+1S)
Capacity	100 m ³ / hr
Make	Akash
Head	4 MWC
Type	Tri lobe roots blower
Capacity	5.0 HP
Motor Make	Crompton Greaves

8.Acoustic Enclosure

DESCRIPTION	
Type	BOOTH
MOC	18swg CRCA sheets, EPDM gaskets
Acoustic Material	Rock wool of 100mm thickness 40Kg/m3 Density
Provision	Must include walk - in acoustic hood
Surface treatment	Powder coated
Size	6feet Length x 8feet Width x 6feet Ht





9. Coarse Bubble Membrane Aerator

DESCRIPTION	
Quantity	10 Nos
Flow	3.8 m ³ /hr
Make	Scogen
Size	225 mm Dia
MOC	EPDM
Medium Of Flow	Air
Type	Disc Type
Dimension Of Bubble	5mm
Temperature Tolerance	85°C
Type Of Arrangement	Retrievable

10. Fine Bubble Membrane Aerator

DESCRIPTION	
Quantity	14 Nos
Flow	5.5 m ³ /hr
Make	Scogen
Size	300 mm Dia
MOC	EPDM
Medium Of Flow	Air
Type	Disc Type
Dimension Of Bubble	5mm
Temperature Tolerance	85°C
Type Of Arrangement	Retrievable

11.MBR Permeate Pump

DESCRIPTION	
Quantity	2 Nos (1W+1S)
Motor	1HP
Head	20m
Type	Monobloc Pump (SS Wetted Parts)
Make	Kirloskar KSMB 129



12. MBR Membrane

DESCRIPTION	
Quantity	200 Sq.M
Make	Hollowbrane
Moc Of Skid	SS 304
Type	Submerged hollow fibre membrane
Pore Size	0.06 Micron
Membrane Construction	R-PVDF
Air Scoring	Inbuild in SS Skid (Coarse)
DESCRIPTION	

13. Membrane Lifting hook

DESCRIPTION	
Quantity	1Set
Make	IAPL
MOC	MS - Bullet Bolt
(Along with chain pulley block and chain for lifting of membrane stand)	

14. Back Wash Pump

DESCRIPTION	
Quantity	2 Nos
Motor	1 HP
Head	20 m
Type	Monobloc Pump (SS Wetted Parts)
Make	Kirloskar KSMB 129

15. RAS Pump

DESCRIPTION	
Quantity	2 Nos
Head	10 M
Flow	2.0 M ³ / Hr
Recirculation Cycle	3 Times/ Day
Motor	1.0 HP
Type	Monoblock
Make	Kirloskar

16. Electromagnetic Flow Meter

DESCRIPTION	
Quantity	2 Nos
Size	1"
Make	Aster
MOC	Cast Aluminum enclosure, Flange MoC: Mild Steel, Electrode MoC: SS316,



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17. UV System

DESCRIPTION	
Quantity	1No
Model	SS304, Tube type
Capacity	0-3000LPH
Make	Alfa
Power	0.04

18. Motorised Valve

DESCRIPTION	
Interconnected Pipes, Fittings and Valves	2 Nos
MOC	CI
Type	Electrical with handle

19. Piping

DESCRIPTION	
Interconnected Pipes, Fittings and Valves	1 Lot
Make UPVC and CPVC	Supreme/ Astral
MOC	UPVC SCH 40 and Fittings SCH 80
MOC	MS - C - Class -Header line to Airline flange
MOC	CPVC - Airline Flange to Delivery
Make MS	TATA/JINDAL
Pressure Gauge	Online

20. Electrical Works and Fittings

DESCRIPTION	
Interconnected Electrical Pipes, Fittings	1 Lot
Type	Flexible Type cables
Cable trays	Perforated cable trays
Earthing	Complete earthing and distribution works





21. Chemical Dosing System with Agitator

DESCRIPTION	
Quantity	3 Nos
Type	Diaphragm
Capacity	0-6LPH
Pressure	4 Bar
End Connection	4/6mm
Max. Suction Pressure	1.5m
Make	Infinity
PP -Storage Tank	100 Liters

22. Sludge Transfer Pump

DESCRIPTION	
Quantity	2 Nos (1W+1S)
Motor	0.5 HP
Head	10 m
Type	Submersible Pump
Make	Kirloskar

23. Online PH Meter

DESCRIPTION	
Quantity	1 No
Make	Aster
Specification	0-12





24. Sludge Dewatering Machine

ITEM	SPECIFICATION
Project Type	Sludge Handling system
FLOW RATE	0.5 M3/HR
Type of Industry	STP- 40-200 kld plant
OPERATING TEM	40DEG
Sludge Type	Primary + Secondary Sludge
Sludge Concentration	10gms/liter (1%)
Dimensions (L × W × H)	1200 mm × 500 mm × 600 mm
Total Power	0.5 HP
Power Supply	440 V / 50 Hz, 3 Phase
Thickening cum Material	SS304 in thickening zone and SS304 in pressure zone
Compaction Cylinder Quantity	One
Model	Standard TEFC SCR Induction Motor
Make	Bharat Bijlee / Rotomotive
Screw Motor Power	0.25 HP
Quantity	One
Operation	Controlled by VFD
Dimensions (Motor Unit)	430 mm × 430 mm × 550 mm
Flocculation Tank Capacity	100 Liters
Flocculation Tank Material	SS 304
Flocculation Tank Motor Model	Standard TEFC SCR Induction Motor
Flocculation Tank Motor Make	Bharat Bijlee / Rotomotive
Agitation Motor Power	0.25 HP
Agitation Motor Quantity	One
Operation	Controlled by VFD
Screw Gear Box Type	Heliworm
Screw Gear Box Make	Power Build
Flocculator Gear Box Type	Worm Reducer 25:1
Flocculator Gear Box Make	Rotomotive / Motovario



Electrical Load Details: -

S. No	Description	Operating Load KW	Connected Load KW	Quantity	Running Hrs.	Total KWH/Day
1.	Raw Water Pump (Pre-Collection to Flocculation Chamber)	0.375	0.75	2Nos	20.0	7.5
2.	Raw Water Pump (Collection Tank to Anoxic Tank)	0.75	1.5	2 Nos	20.0	15.0
3.	MBR Feed Pump	0.75	1.5	2Nos	20.0	15.0
4.	Back Wash Pump	0.75	1.5	2Nos	1.0	0.75
5.	Air Blower with Motor	3.75	7.5	2Nos	24.0	90.0
6.	RAS Pump	0.75	0.75	2 Nos	1.0	0.75
7.	UV System	0.04	0.04	1No	20.0	0.8
8.	Sludge Transfer pump	0.375	0.75	2Nos	1.0	0.375
9.	Chemical Dosing systems	0.04	0.12	3Nos	20.0	2.4
10.	Agitator For Anoxic Tank	1.1	1.1	1 No	20.0	22.0
11.	Mechanical Screen Chamber	1.5	1.5	1 No	20.0	30.0
12.	Oil Skimmer	0.375	0.75	2 Nos	20.0	15.0
13.	Screw press	0.375	0.375	1 No	1.0	0.375
TOTAL		10.93	18.135			199.95
70% of Total Load						139.9

OPERATING COST

S. No	Description	Cost per day
1.	Power consumption cost per day	Rs. 1399(At Rs. 10 per unit)
2.	Chemical And Consumables per day	Rs 100
3.	Total Cost per day	Rs. 1499
4.	Operating cost for Treating 50,000litres Sewage per day	= 1499/50000 Rs. 0.029.9/ Liter Rs. 29.9/ 1000 L



Civil and MS Dimensions

S.No	Description	Size (M)	Quantity	Type	Scope	Volume m ³
1.	SCREEN CHAMBER	2.3 m X 1.1 m x 0.6m T.D	1No	RCC	Client	1.5
2.	OIL TRAP CHAMBER	2.3 m X 1.7 m x 0.6 T.D	1No	RCC	Client	2.4
3.	PRE-COLLECTION TANK	3.7 m X 2.5 m X 1.4m T.D	1No	RCC	Client	10.6
4.	FLOCULATION TANK	2.0 m X 1.2 m X 0.3m T.D	1No	RCC	Client	0.72
5.	PRIMARY SETTLING TANK	1.2 m X 3.1 m X 1.4m T.D	1No	RCC	Client	4.3
6.	COLLECTION TANK	4.2 m X 3.5 m X 2.2m T.D	1No	RCC	Client	28.1
7.	ANOXIC TANK	4.2 m X 2.3 m X 2.2 m T.D	1No	RCC	Client	11.8
8.	AERATION TANK	2.6 m X 4.8 m X 2.2 m T.D	1No	RCC	Client	21.4
9.	MBR TANK	2.6 m X 2.1 m X 2.2 m T.D	1No	RCC	Client	10.3
Total Volume						91.1m ³

Special Note:

If the client opts for RCC construction, all civil works shall be under the client's scope.



COMMERCIAL OFFER

S. No	Description	Amount (Rs.)
1.	The Price for Supply, Installation and Commissioning of Conventional Sewage Treatment Plant Capacity of 50 KLD MBR – Hollowbrane Membrane	23,14,000.00
2.	The Price for Supply, Installation of OCEMS – 1 No Make: Aster	3,28,600.00
Total		26,42,600.00
GST 18%		4,75,668.00
Grand Total		31,18,268.00
INR THIRTY – ONE LAKHS EIGHTEEN THOUSAND TWO HUNDRED AND SIXTY - EIGHT ONLY		

Terms and Conditions: -

Delivery	: Within 6 to 8 weeks from the date of work order along with Advance Payment.
GST	: 18% Included in the above price.
Erection & Comm.	: Included in the above price.
Freight	: Included in the above price
Unloading	: Client Scope
Validity	: 20 days.

Warranty: -

12 months from the date of Erection or 18 months from the date of supply whichever is earlier. This warranty includes only inherent manufacturing defects; it does not cover any replacement for normal wear and tear.

Payment Terms: -

- ❖ Mobilization Advance: 50% of the total project cost upon receipt of the Purchase Order.
- ❖ Material Supply: 40% of the total project cost, along with 18% GST, to be paid against Submission of a proforma invoice, prior to the supply of materials.
- ❖ Handover Payment: 10% of the total project cost against handover on an immediate basis.
- ❖ In case of delays due to electrical supply or insufficient feed water, ₹25,000 can be kept on Hold and the remaining payment should be released upon completion of work.
- ❖ Balance payment exceeding 45days from tax invoice is subjected to fall within MSME Norms and compliance.





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SCOPE OF EXCLUSIONS: -

1. Water, Panel Earthing & Chemicals required for erection and commissioning are to be provide by client.
2. Civil Works, Plant Room & Shed, Treated water Pump.
3. Raw Sewage inlet pipe to Screen Chamber, Interconnecting Pipes in the Tanks and Treated water final discharge pipe line are not included in our scope of supply.
4. Air vent line for all the treatment tanks.
5. Electrical Cabling will be laid from our Panel Board and if the Panel Board is above 5m from the proposed plant it will cost extra.
6. Lighting arrangements and generators if required during erection work, Ladder, Handrails and Walkway arrangement for tanks
7. Providing incoming power supply to the proposed panel board (3-phase, 415V, 50Hz), including proper panel earthing, single-phase preventer, and necessary plug points and switches
8. During Commissioning we will depute our Engineer for the same and we require an Operator to operate the plant.
9. Any other things which are not mentioned apart from our proposal.

Note:

1. After supply of materials to site, client is the responsible if there is any problem occurred due to mishandling of materials by you.
2. All the lump sum electrical / plumbing material will be taken back by the supplier after

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